Hashing

$$a \quad b \quad c$$
$$0 = 10^2 \times 07 + 10^1 \times 08 + 10^0 \times 09$$

$$b \quad a \quad c$$
$$a \quad b$$
$$10^2 \times 07 + 10^1 \times 08 + 10^0 \times 09$$

Load factor $\lambda = \frac{N}{T.S} \rightarrow \frac{\text{total item}}{\text{Table size}}$

$\therefore \lambda > 0.5 \rightarrow$ rehashing in backend
↓
transferring in another table.

open Address

$h(x) = [\text{key} + f(i)] \% t_s$

$f(i)$ → conflict taking help of another.
t_s → tablse

Key % size
(K+1) % size
2 "

↓ linear probing



Why free space is guranted in linear prob but not in quadratic

OOP 1

Seperate chaining worst case O(n+1)
all are in a single linked list.

Mark Allen — Ch2:5.

Rabin Karp

N S_1 → abcdecc adef

M S_2 → bcd

for (si)

for ( )

|

O(MN)

↳ instead

O(M+N) → Rabin Karp.

hash (S_2)

hash (window)

S[i] $\times 10^0$ + S[ ] $\times 10^1$ + S[ ] $\times 10^2$ ----

---- ? $\times 10^9$

Again becomes O(N×M)

Rolling hash func

Rolling hash

$$abcd = a\times10^0 + \boxed{b\times10^1 + c\times10^2 + d\times10^3}$$

hash

reuse the hash value

divide by base

give

$$bcde = b\times10^0 + c\times10^1 + d\times10^2 + e\times10^3$$

$$H_i = (H_{i-1} - S_{i-1}) / b + S_i \times b^{m-1}$$

$$O(N\times N)$$

Next → Trie.